SENSOR

NASPIER V2
IMU BOARD

Dosya: SENSOR_IMU.kicad_sch

HIROSE 34P
CONNECTOR

FLEX_CABLE

Dosya: FLEX_CABLE_sch.kicad_sch

CARRIER

PAB STANDARD
CARRIER BOARD

Dosya: CARRIER.kicad_sch

CORE

Dosya: CORE.kicad_sch

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	NASPIER CORE: SYSTEM (CORE / SENSOR / CARRIER)
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 1 Total 16
		REV0 VERO	A3	VAP-SAN

HIROSE 34P
FFC CONNECTOR

CORE_H743

STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)

Dosya: CORE_H743.kicad_sch

CORE_POWER

IMU BUSSES 3.3V
REGULATORS

FC & AUX 3.3V
REGULATORS

Dosya: CORE_POWER.kicad_sch

CORE_ONBOARD

AT24C64D
EEPROM-I2C

FRAM
SPI

ICM-45686
IMU-SPI

MS5611
BARO-I2C

Dosya: CORE_ONBOARD.kicad_sch

CORE_CONNECTORS

HIROSE 100P
CONNECTOR

HIROSE 50P
CONNECTOR

Dosya: CORE_CONNECTORS.kicad_sch

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	CORE BOARD
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version	Size	Page 2 Total 16
		REV0 VER0	A4	VAP-SAN

U1A
STM32H743IIK6

U1B
STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)
NON-STANDARD PIXHAWK CHIP

FMU 3V3 DIGITAL LINE CAPS

FMU 3V3 ANALOG LINE CAPS

R-G-B LEDS

CRYSTALS

SPI4 EXPANSION

REV SENS
DO NOT PLACE

RTC BATTERY LINE

C300 DNP: VBAT decoupling for testing only.
MLCC leakage drains the coin cell.

DEBUG INTERFACE

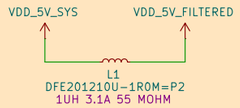
THIS CONNECTOR JUST FOR
DEBUG/PROGRAMMING
POINTS GROUPS

PWR_FLAG (ERC drivers for the rails on this sheet)

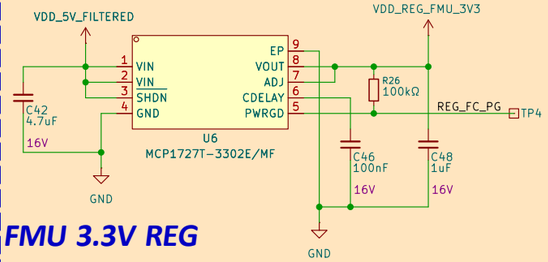
APPROVED (C8, 2026-10-25): Y1 16 MHz CL 18 pF -> C9 = C15 = 2 x (18 - 4) = 28 -> 27 pF C0G; Cstray 4 pF.
APPROVED (C8, 2026-10-25): Y2 32.768 kHz CL 7 pF -> C20 = C27 = 2 x (7 - 3.5) = 7 -> 6.8 pF C0G; verify LSE start-up.
APPROVED (B-SMALL, 2026-10-27): Y2 Abracon ABS07-32.768KHZ-7-T, 3.2 x 1.5 mm (smaller than 3.2 x 2.5, same CL 7 pF / 20 ppm).
PCB: check the land pattern against the ABS07 drawing.

Schematic	Naspier Core	Create at	2026-09-20
Board	NASPIER_CORE_REV0_VER0	Update at	2026-10-15
Drawn	Ahmet Vapurcu	Page	CORE_H743
Reviewed			
	Naspier Core		
	Version: REV0 VER0	Size: A3	Page 3 Total 16
			VAP-SAN

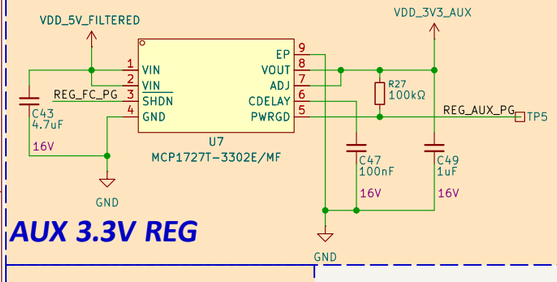
5V LINE FILTER



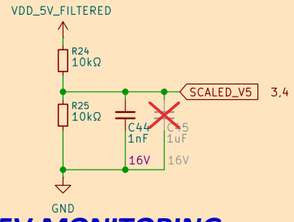
FMU 3.3V REG



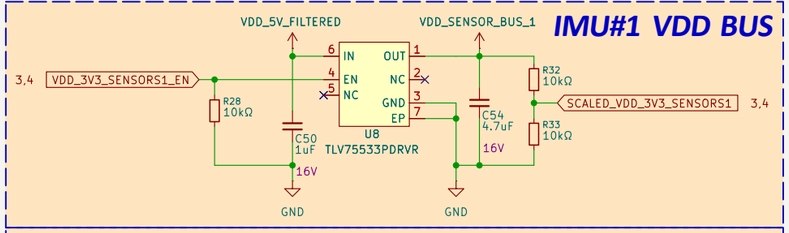
AUX 3.3V REG



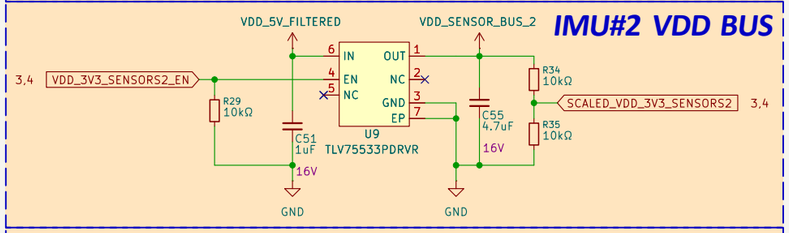
5V MONITORING



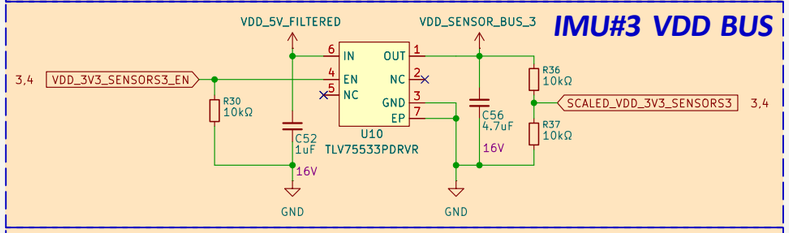
IMU#1 VDD BUS



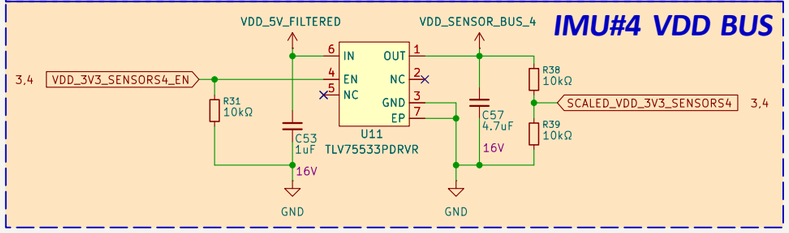
IMU#2 VDD BUS



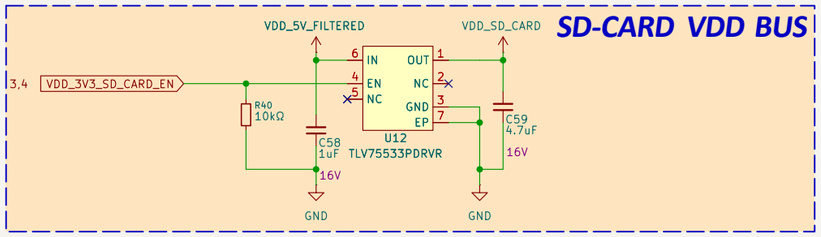
IMU#3 VDD BUS



IMU#4 VDD BUS

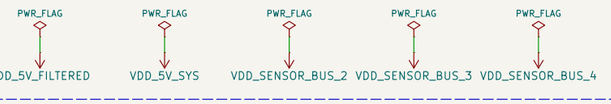


SD-CARD VDD BUS



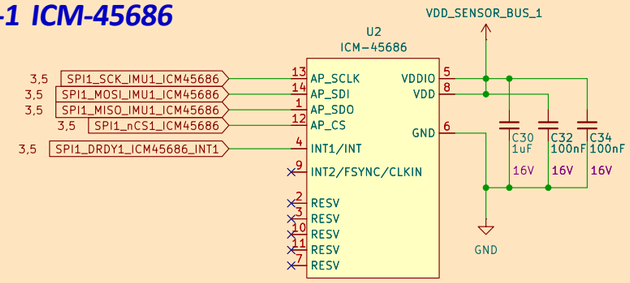
NOT APPROVED (B-LDO, 2026-10-25): LDL212PV33R -> TLV75533PDRVR, WSON-6 2x2 (TI DRV).
Pins: 1 OUT, 2 NC, 3 GND, 4 EN, 5 NC, 6 IN, 7 pad. Pins 2 / 5 float (no-connect flags).
VIN 5.5 V recommended, 6 V absolute max (LDL212: 18 V): 5 V rail spikes (hot plug, servo back-feed) can exceed it. To check later: TVS or LC filter at IN if needed. 500 mA max, no sense pin.

PWR_FLAG (ERC drivers for the rails on this sheet)

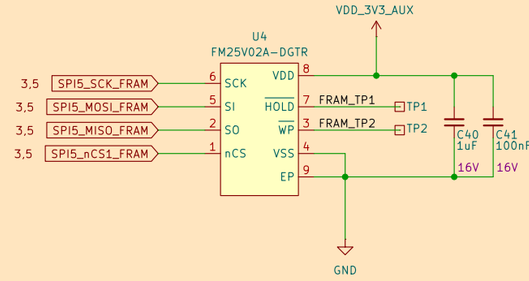


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CORE_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 4 Total 16
		REV0 VER0	A3 / VAP-SAN

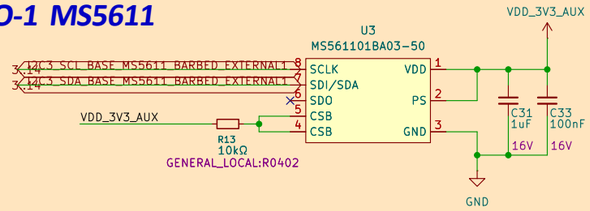
IMU-1 ICM-45686



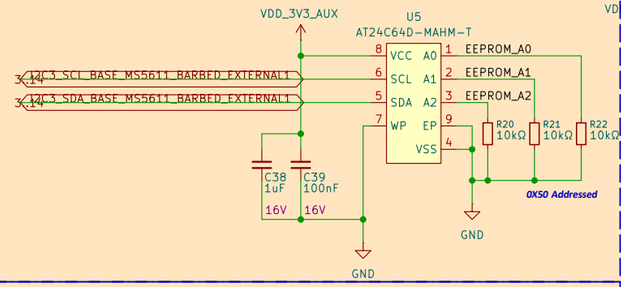
FRAM

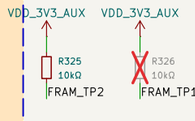


BARO-1 MS5611



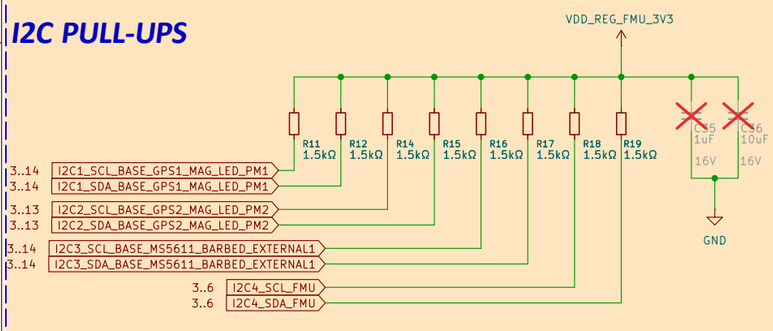
ONBOARD EEPROM



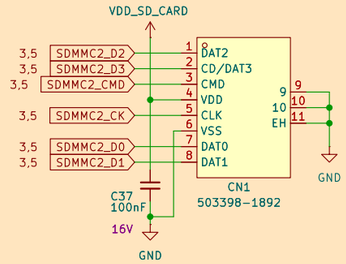


U4 (FM25V02A): -WP (FRAM_TP2) pulled up by R325, writes allowed by default (PX4 parameter storage);
TP2 can still pull it low (with WPEN set, the status register is then write protected).
-HOLD (FRAM_TP1) has a weak internal pull-up, floating is allowed; R326 10 k DNP as an option.

I2C PULL-UPS

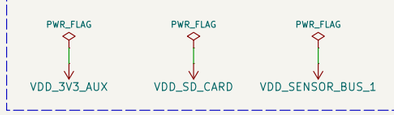


SD-CARD



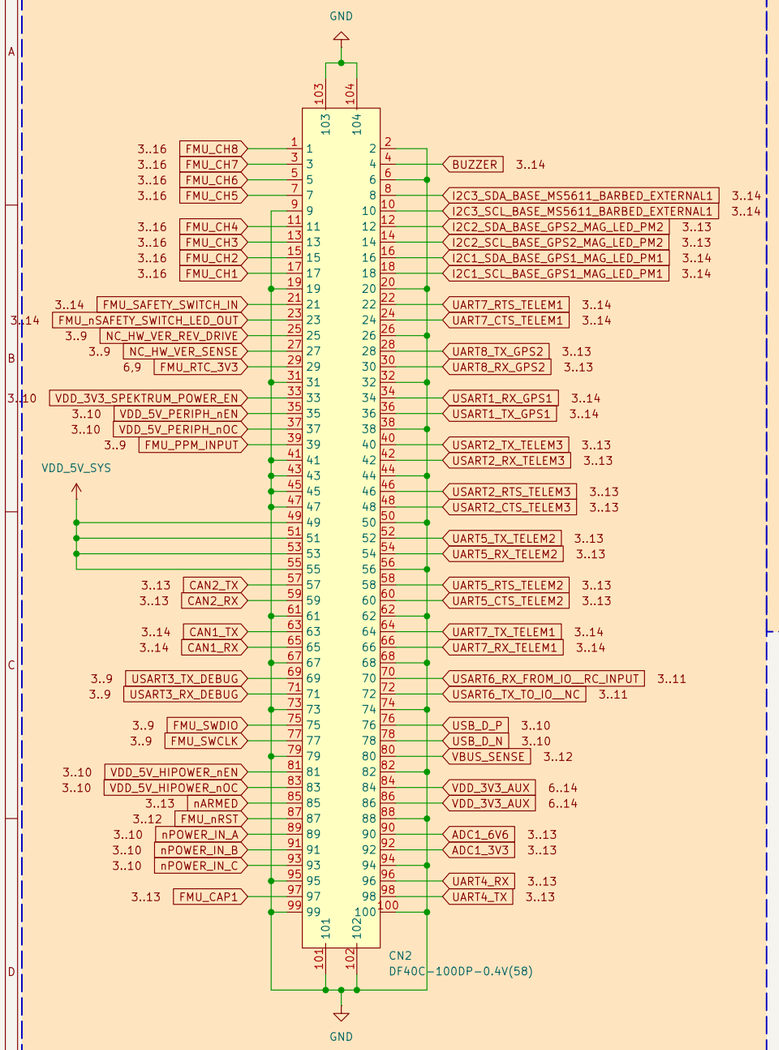
Push-In
Push-Out

PWR_FLAG (ERC drivers for the rails on this sheet)



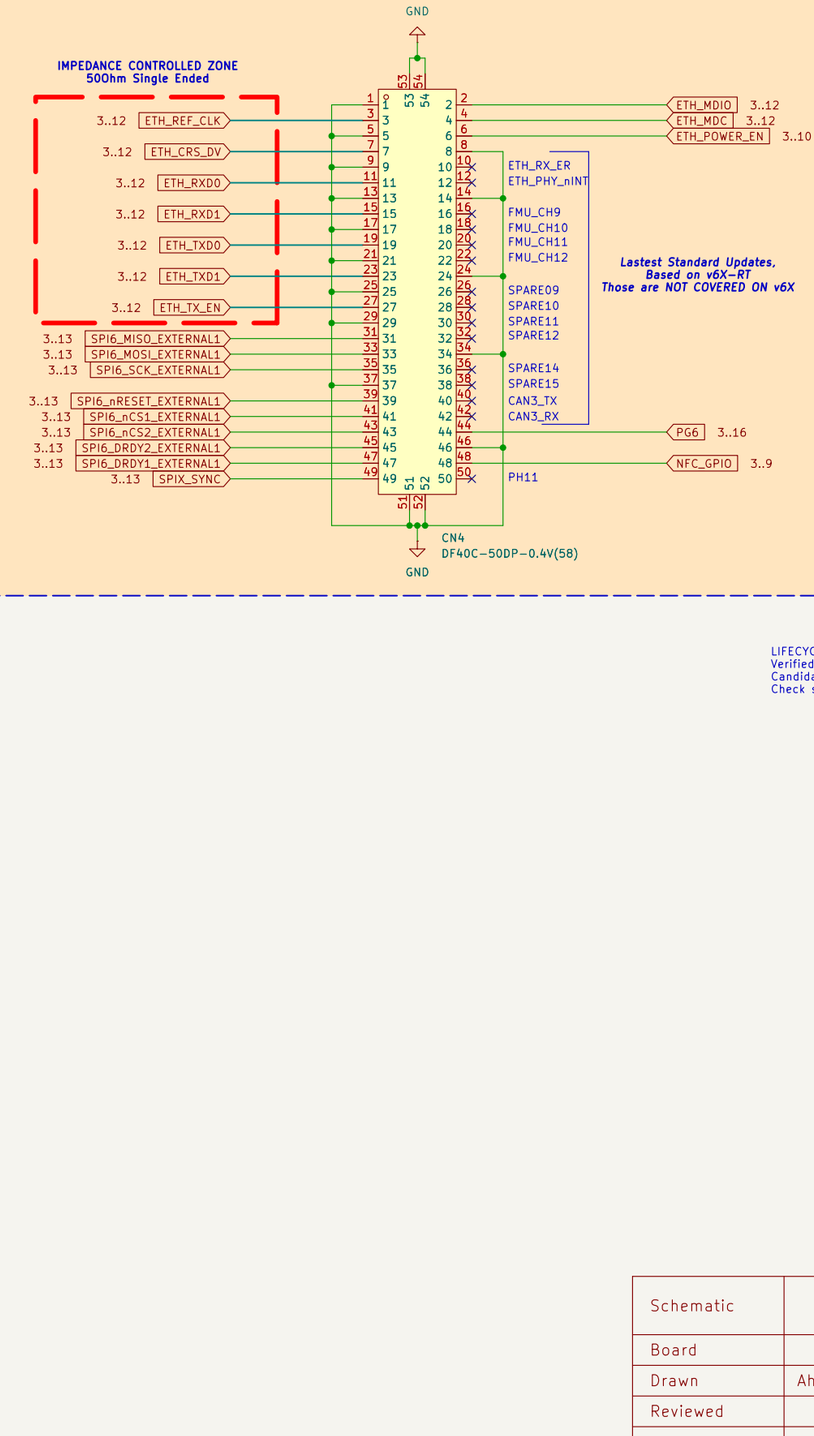
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	**CORE_ELEMENTS**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 5 Total 16
		REV0 VERO	A3 / VAP-SAN

PAB X1 100P HIROSE

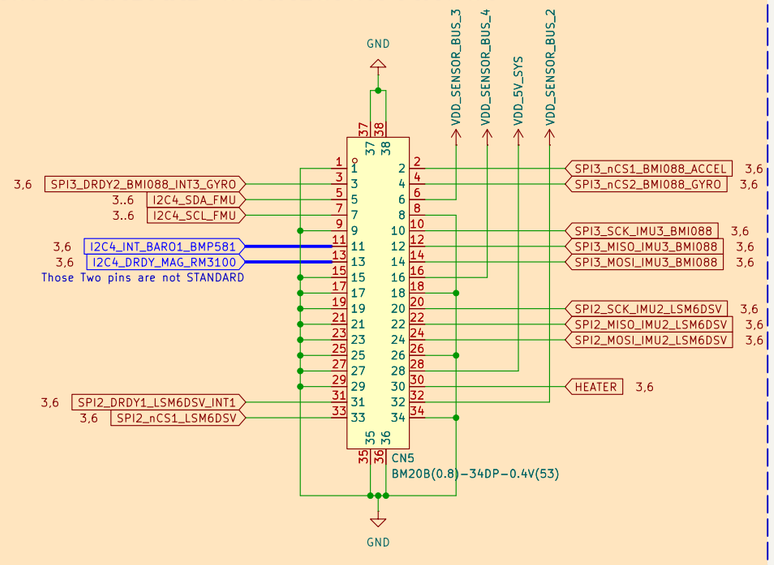


PAB X2 50P HIROSE

v6X-RT only (not on v6X cores): FMU_CH9-CH12, CAN3, ETH_RX_ER, ETH_PHY_nINT, SPARE09-12 / 14 / 15, PH11. Fit / route them only for v6X-RT support.



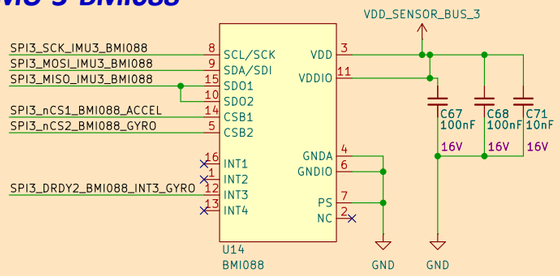
PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT



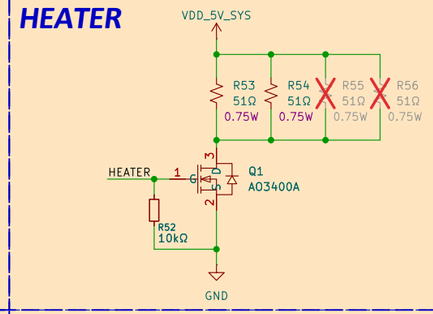
LIFECYCLE (B-BM20B): BM20B(0.8)-34DP last-time-buy 2026-11-30 (DigiKey UK).
Verified 2026-10-25: header (CN5 / CN6) and flex receptacle (CN14 / CN20) change together as a mated pair.
Candidate pairs, 34 pins 0.4 mm: Hirose DF40C-34DP-0.4V(51) + DF40C-34DS-0.4V(51); Panasonic P4S AXT434124 + AXT334124.
Check stack height, pin 1 and pinout of the chosen pair against the flex sheet.

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	**CORE_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 6 Total 16
		REV0 VERO	A3 / VAP-SAN

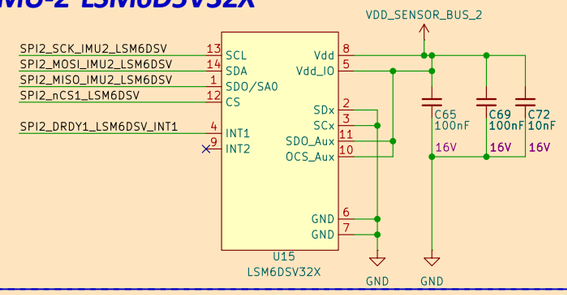
IMU-3 BMI088



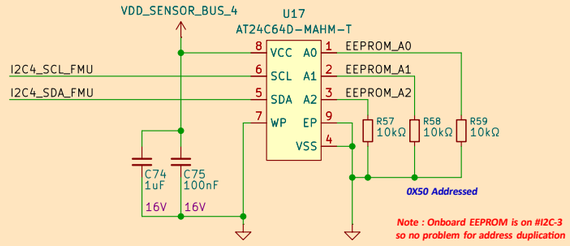
HEATER



IMU-2 LSM6D5V32X

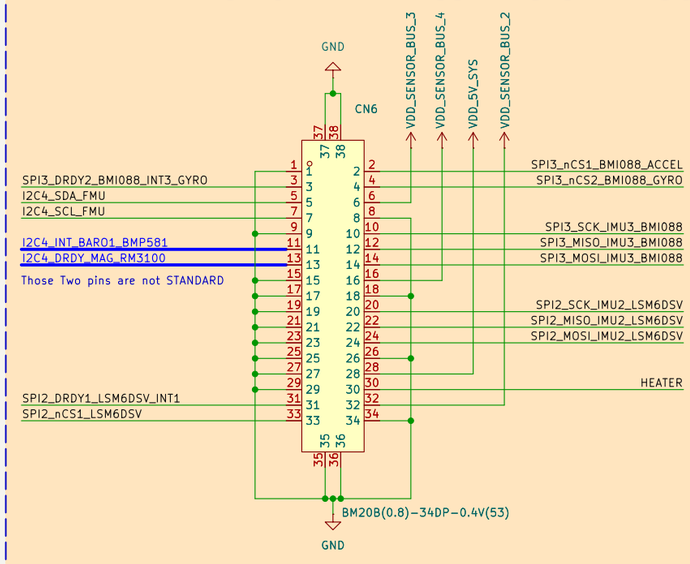


EEPROM



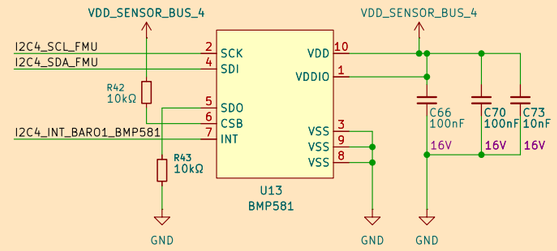
C11: R53–R56 Rohm LTR18EZPF51R0, 0.75 W wide-terminal 1206 (0612 footprint).
R55 / R56 DNP: fit them for more heater power when needed.

PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT

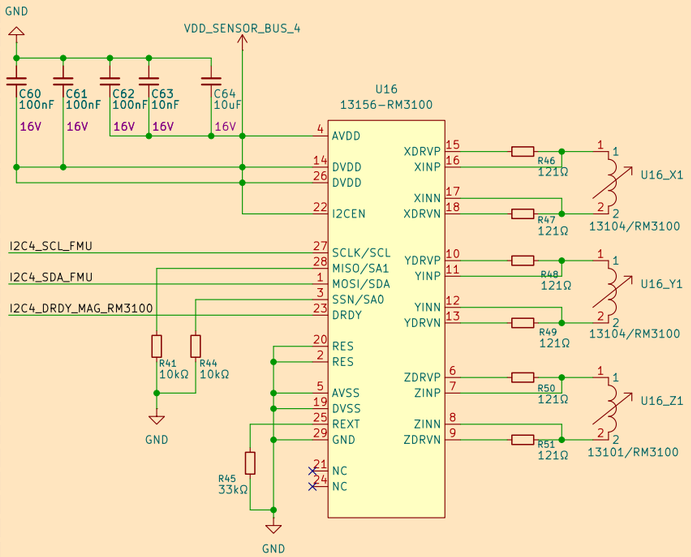


FOR BETTER FFC LAYOUT
PINOUT HERE WILL CHANGE

BARO-2 BMP581



MAG-RM3100



LIFECYCLE (B-BM20B): BM20B(0.8)-34DP last-time-buy 2026-11-30 (DigiKey UK).
Verified 2026-10-25: header (CN5 / CN6) and flex receptacle (CN14 / CN20) change together as a mated pair.
Candidate pairs, 34 pins 0.4 mm: Hirose DF40C-34DP-0.4V(51) + DF40C-34DS-0.4V(51); Panasonic P4S AXT434124 + AXT334124.
Check stack height, pin 1 and pinout of the chosen pair against the flex sheet.

Schematic	**Naspier Core**	Create at	2026-09-20
Board	NASPIER_CORE_REV0_VER0	Update at	2026-10-15
Drawn	Ahmet Vapurcu	Page	**SENSOR_BOARD**
Reviewed			
	Naspier Core		
	Version: REV0 VER0	Size: A3	Page 7 Total 16
			VAP-SAN

CARRIER_B2B

Dosya: CARRIER_B2B.kicad_sch

CARRIER_CONN_1

Dosya: CARRIER_CONN_1.kicad_sch

CARRIER_IO_MCU

Dosya: CARRIER_IO_MCU.kicad_sch

CARRIER_POWER

Dosya: CARRIER_POWER.kicad_sch

CARRIER_CONN_2

Dosya: CARRIER_CONN_2.kicad_sch

CARRIER_ELEMENTS

Dosya: CARRIER_ELEMENTS.kicad_sch

CARRIER_PWM

Dosya: CARRIER_PWM.kicad_sch

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER BOARD**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 8 Total 16
		REV0 VER0	A4	VAP-SAN

PAB X1 100P HIROSE

CN8
DF40HC(3.0)-100DS-0.4V(51)

3..16 FMU_CH8 1
3..16 FMU_CH7 3
3..16 FMU_CH6 5
3..16 FMU_CH5 7
9
3..16 FMU_CH4 11
3..16 FMU_CH3 13
3..16 FMU_CH2 15
3..16 FMU_CH1 17
19
3..14 FMU_SAFETY_SWITCH_IN 21
3..14 FMU_nSAFETY_SWITCH_LED_OUT 23
3..9 NC_HW_VER_REV_DRIVE 25
3..9 NC_HW_VER_SENSE 27
6,9 FMU_RTC_3V3 29
31
3..10 VDD_3V3_SPEKTRUM_POWER_EN 33
3..10 VDD_5V_PERIPH_nEN 35
3..10 VDD_5V_PERIPH_nOC 37
3..9 FMU_PPM_INPUT 39
41
43
45
47
49
51
53
55
3..13 CAN2_TX 57
3..13 CAN2_RX 59
61
3..14 CAN1_TX 63
3..14 CAN1_RX 65
67
3..9 USART3_TX_DEBUG 69
3..9 USART3_RX_DEBUG 71
73
3..9 FMU_SWDIO 75
3..9 FMU_SWCLK 77
79
3..10 VDD_5V_HIPOWER_nEN 81
3..10 VDD_5V_HIPOWER_nOC 83
3..13 nARMED 85
3..12 FMU_nRST 87
3..10 nPOWER_IN_A 89
3..10 nPOWER_IN_B 91
3..10 nPOWER_IN_C 93
95
3..13 FMU_CAP1 97
99

2
4 BUZZER 3..14
6
8 I2C3_SDA_BASE_MS5611_BARBED_EXTERNAL1 3..14
10 I2C3_SCL_BASE_MS5611_BARBED_EXTERNAL1 3..14
12 I2C2_SDA_BASE_GPS2_MAG_LED_PM2 3..13
14 I2C2_SCL_BASE_GPS2_MAG_LED_PM2 3..13
16 I2C1_SDA_BASE_GPS1_MAG_LED_PM1 3..14
18 I2C1_SCL_BASE_GPS1_MAG_LED_PM1 3..14
20
22 UART7_RTS_TELEM1 3..14
24 UART7_CTS_TELEM1 3..14
26
28 UART8_TX_GPS2 3..13
30 UART8_RX_GPS2 3..13
32
34 USART1_RX_GPS1 3..14
36 USART1_TX_GPS1 3..14
38
40 USART2_TX_TELEM3 3..13
42 USART2_RX_TELEM3 3..13
44
46 USART2_RTS_TELEM3 3..13
48 USART2_CTS_TELEM3 3..13
50
52 UART5_TX_TELEM2 3..13
54 UART5_RX_TELEM2 3..13
56
58 UART5_RTS_TELEM2 3..13
60 UART5_CTS_TELEM2 3..13
62
64 UART7_TX_TELEM1 3..14
66 UART7_RX_TELEM1 3..14
68
70 USART6_RX_FROM_IO__RC_INPUT 3..11
72 USART6_TX_TO_IO__NC 3..11
74
76 USB_D_P 3..10
78 USB_D_N 3..10
80 VBUS_SENSE 3..12
82
84 VDD_3V3_AUX 6..14
86 VDD_3V3_AUX 6..14
88
90 ADC1_6V6 3..13
92 ADC1_3V3 3..13
94
96 UART4_RX 3..13
98 UART4_TX 3..13
100

VDD_5V_SYS

VDD_3V3_AUX
R327 NC
NC_HW_VER_SENSE
R328 NC
NC_HW_VER_REV_DRIVE

GND

MountingHole M2 x4
CORE SCREWED STANDOFFS

H201 H202 H203 H204

GND

HW_VER_SENSE divider (RC05 page 2, R13 / R18): R327 from FMU_VDD_3V3 (X1 84 / 86 = VDD_3V3_AUX), R328 to HW_VER_REV_DRIVE. Both DNP; values set the baseboard version. Net names keep the NC_ prefix.

PAB X2 50P HIROSE

IMPEDANCE CONTROLLED ZONE
50Ohm Single Ended

CN9
DF40HC(3.0)-50DS-0.4V(51)

3..12 ETH_REF_CLK 3
3..12 ETH_CRS_DV 7
3..12 ETH_RXD0 9
3..12 ETH_RXD1 11
3..12 ETH_TXD0 15
3..12 ETH_TXD1 19
3..12 ETH_TX_EN 23
25
3..13 SPI6_MISO_EXTERNAL1 27
3..13 SPI6_MOSI_EXTERNAL1 29
3..13 SPI6_SCK_EXTERNAL1 31
33
3..13 SPI6_nRESET_EXTERNAL1 35
3..13 SPI6_nCS1_EXTERNAL1 37
3..13 SPI6_nCS2_EXTERNAL1 39
3..13 SPI6_DRDY2_EXTERNAL1 41
3..13 SPI6_DRDY1_EXTERNAL1 43
3..13 SPIX_SYNC 45
47
49

2 ETH_MDIO 3..12
4 ETH_MDC 3..12
6 ETH_POWER_EN 3..10
8
10 ETH_RX_ER
12 ETH_PHY_nINT
14
16 FMU_CH9
18 FMU_CH10
20 FMU_CH11
22 FMU_CH12
24
26 SPARE09
28 SPARE10
30 SPARE11
32 SPARE12
34
36 SPARE14
38 SPARE15
40 CAN3_TX
42 CAN3_RX
44 PG6 3..16
46
48 NFC_GPIO 3..9
50 PH11 9

Lastest Standard Updates,
Based on v6X-RT
Those are NOT COVERED ON v6X

GND

v6X-RT only (not on v6X cores): FMU_CH9-CH12, CAN3, ETH_RX_ER, ETH_PHY_nINT, SPARE09-12 / 14 / 15, PH11. Fit / route them only for v6X-RT support.

FMU DEBUG

VDD_3V3_AUX

D34
PGB1010402KR

3..9 USART3_TX_DEBUG
3..9 USART3_RX_DEBUG
D201 BAS70W-7-F DBG_FMU_RX
3..9 FMU_SWDIO
3..9 FMU_SWCLK
D202
ESD7104MUTAG

3..13 SPI6_SCK_EXTERNAL1
3..9 NFC_GPIO
9 PH11
3..12 FMU_nRST
D215
ESD7104MUTAG

CN201
SM10B-SRSS-TB
FMU debug

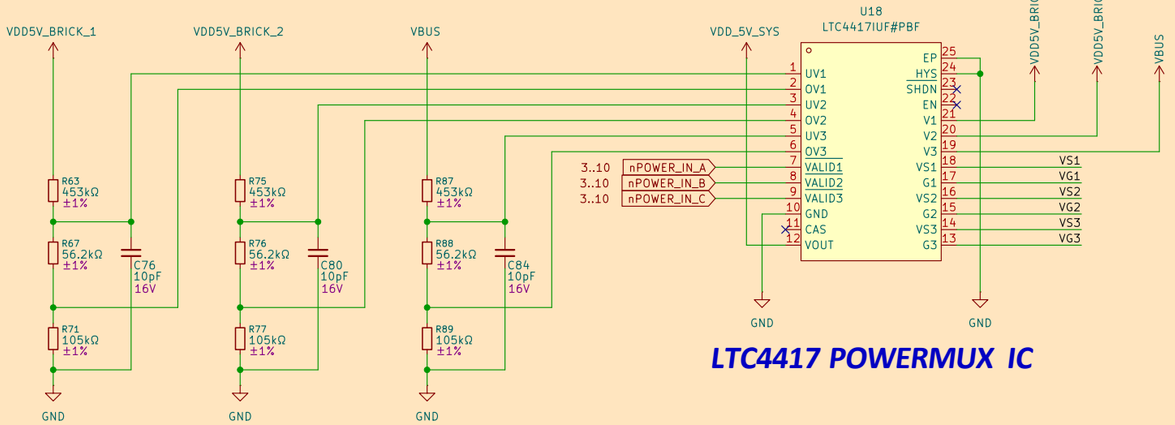
GND

GND

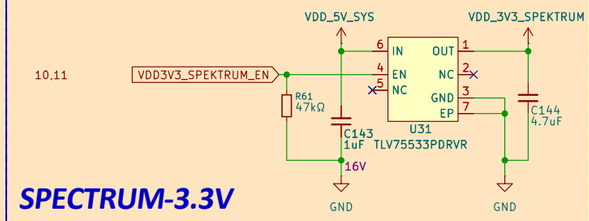
note for using with core module: for FC chip flashing you must use 5V line
this 3.3V just for checking system health
or programmer sense
Other Standard module's fc chip can be powered by this bus

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version: REV0 VER0	Size: A3
		Page 9 Total 16	VAP-SAN

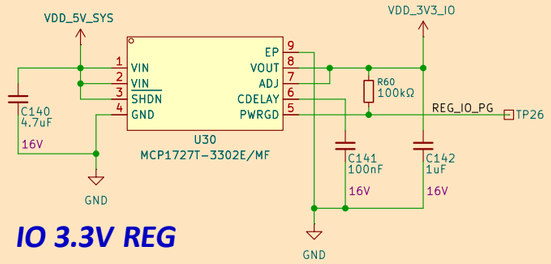
LTC4417 POWERMUX IC



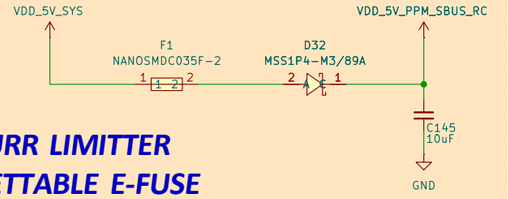
SPECTRUM-3.3V



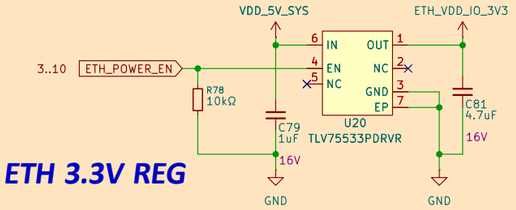
IO 3.3V REG



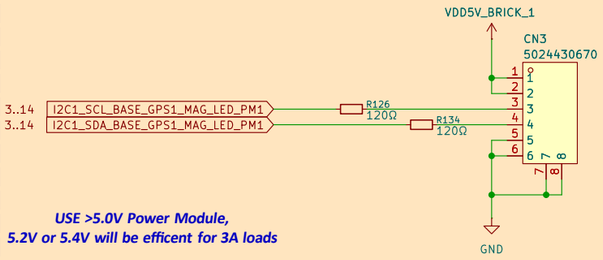
CURR LIMITTER RESETTABLE E-FUSE



ETH 3.3V REG

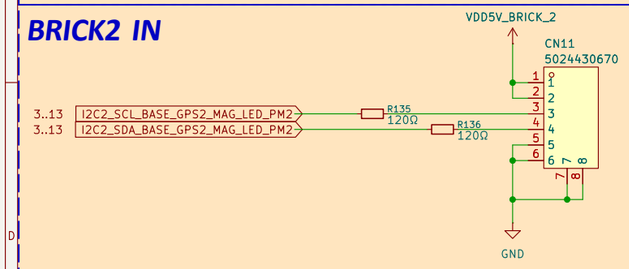


BRICK1 IN

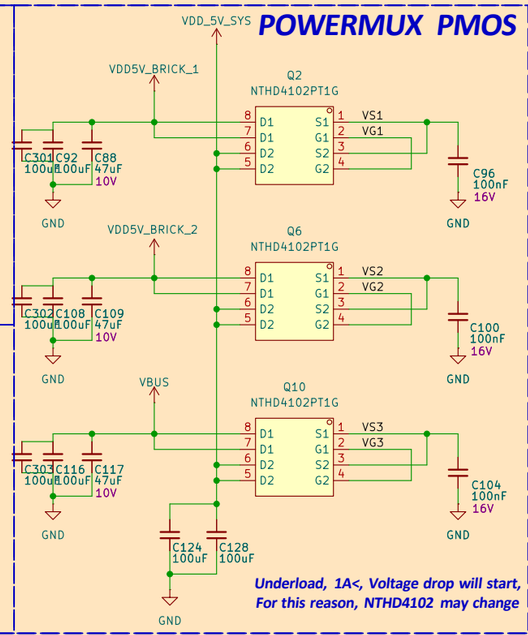


USE >5.0V Power Module,
5.2V or 5.4V will be efficent for 3A loads

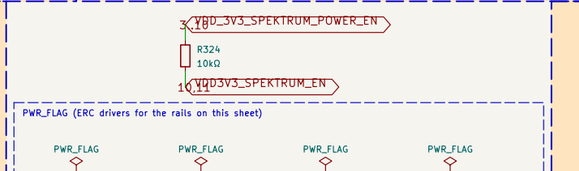
BRICK2 IN



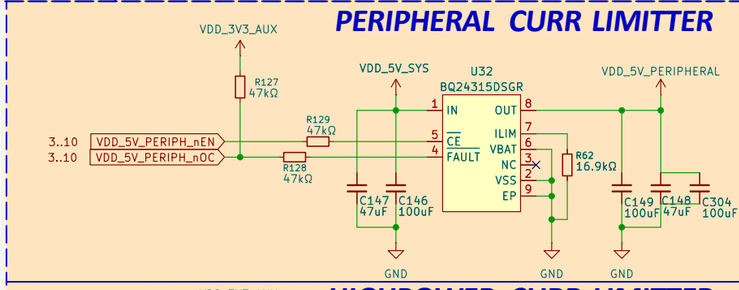
POWERMUX PMOS



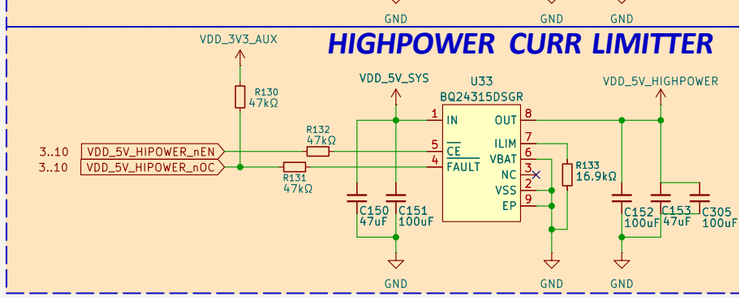
Underload, 1A<, Voltage drop will start,
For this reason, NTHD4102 may change



PERIPHERAL CURR LIMITTER



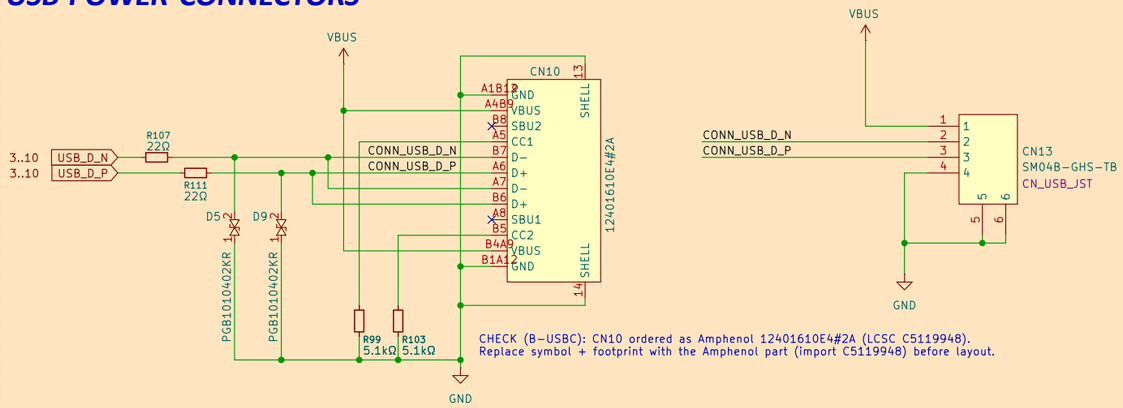
HIGHPOWER CURR LIMITTER



NOT APPROVED (B-LDO, 2026-10-25): LDL212PV33R -> TLV75533PDRVR, WSON-6 2x2 (TI DRV).
Pins: 1 OUT, 2 NC, 3 GND, 4 EN, 5 NC, 6 IN, 7 pad. Pins 2 / 5 float (no-connect flags).
VIN 5.5 V recommended, 6 V absolute max (LDL212: 18 V): 5 V rail spikes (hot plug, servo back-feed) can exceed it. To check later: TVS or LC filter at IN if needed. 500 mA max, no sense pin.

L6 (RC05 page 8): U31 EN = VDD3V3_SPEKTRUM_EN, driven by the IO MCU (PC13) directly and by the FMU (PH2, VDD_3V3_SPEKTRUM_POWER_EN, DS-012) through R324 10 k; R61 47 k pull-down.
Both are the regulator enable (the receiver has no chip enable): the IO MCU can power-cycle it for DSM bind.

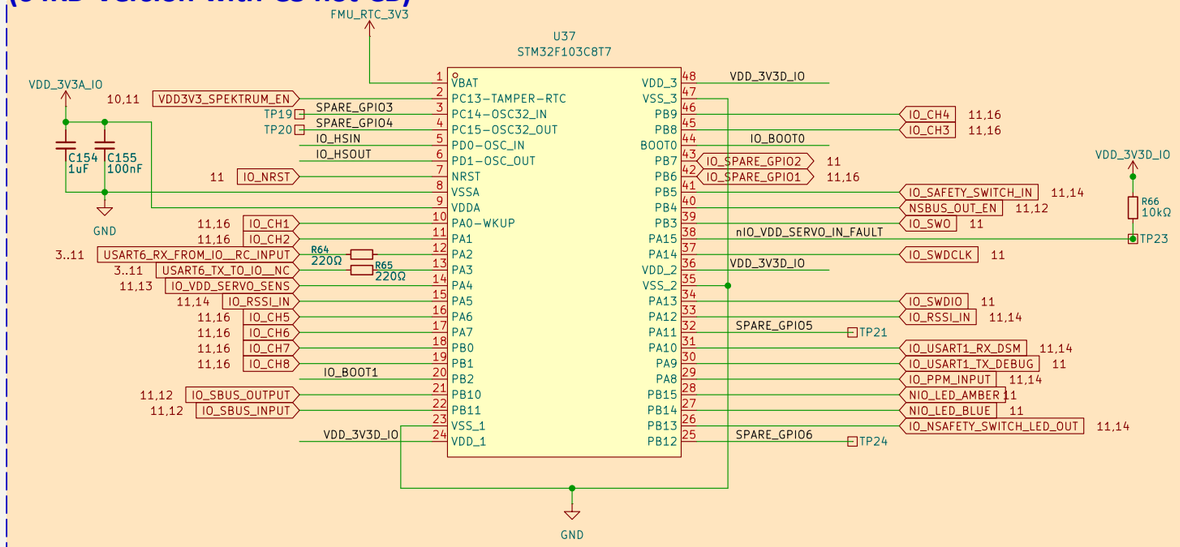
USB POWER CONNECTORS



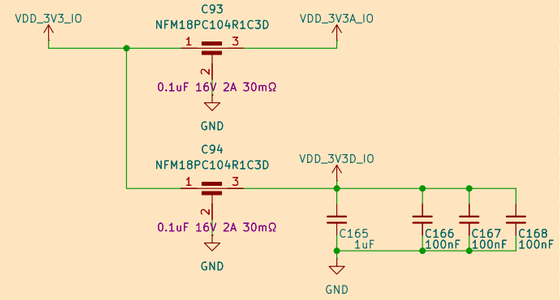
CHECK (B-USBC): CN10 ordered as Amphenol 12401610E4#2A (LCSC C5119948).
Replace symbol + footprint with the Amphenol part (import C5119948) before layout.

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	**CARRIER_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 10 Total 16	
		REV0 VERO / A3 / VAP-SAN	

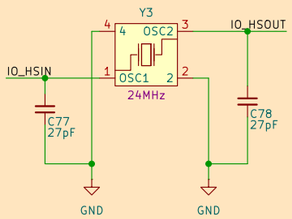
IO F103 CHIP
(64KB Version with C8 not CB)



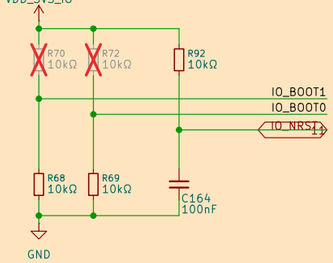
IO POWER FILTER

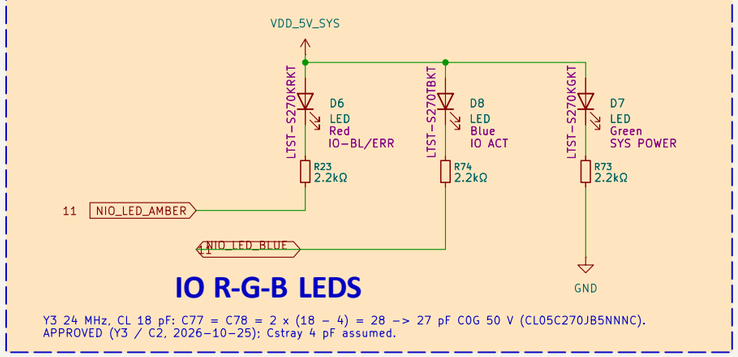


IO CRYSTAL



IO BOOT-NRST

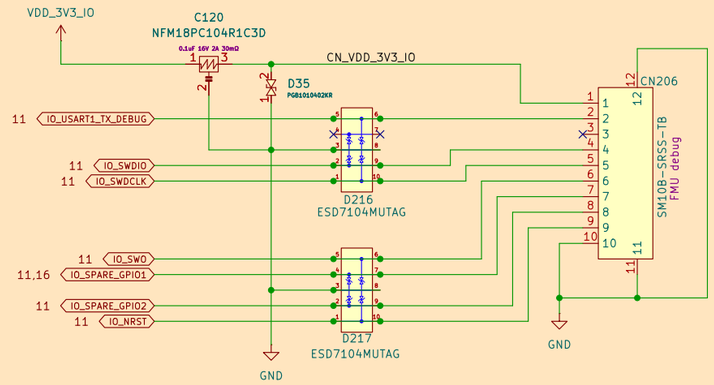




IO R-G-B LEDS

Y3 24 MHz, CL 18 pF: C77 = C78 = 2 x (18 − 4) = 28 -> 27 pF C0G 50 V (CL05C270JB5NNNC).
APPROVED (Y3 / C2, 2026-10-25): Cstray 4 pF assumed.

IO DEBUG



PWR_FLAG (ERC drivers for the rails on this sheet)



Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_IO_MCU**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 11 Total 16
		REV0 VER0	A3	VAP−SAN

CHECK (C17): RC05 drives ETH_NRESET from the MIC5332 POR output.
Here nRST has only R116 (10 k pull-up): the PHY relies on its internal POR.

v6X-RT only: ETH_RX_ER (X2 pin 10) and R112 are used with v6X-RT cores, not with v6X.

ETH_VDD_IO_3V3

C159 10uF, C158 100nF, C157 100nF

C160 100nF, C161 10nF

R114 1.5kΩ, R116 10kΩ

ETH_MDIO, ETH_NRST

C162 1uF, C163 470pF

ETH_VDDCR

U19
LAN8742AI-CZ

VDD2A, VDD1A, VDDIO, VDDCR

RXER/PHYAD0, CRS_DV/MODE2, MDIO, MDC, TXEN, nRST, TXD1, TXD0, RXD1/MODE1, RXD0/MODE0, nINT/REFCLKO

RXN, RXP, TXN, TXP, LED2/nINT/nPME/nINTSEL, LED1/nINT/nPME/REGOFF, XTAL2, XTAL1/CLKIN, RBIAS, EPAD

ETH_RX_ER, R112
3..12 ETH_CRS_DV, R113 22Ω
3..12 ETH_MDIO
3..12 ETH_MDC
3..12 ETH_TX_EN
ETH_NRST
3..12 ETH_TXD1
3..12 ETH_TXD0
3..12 ETH_RXD1, R117 22Ω
3..12 ETH_RXD0, R118 22Ω
3..12 ETH_REF_CLK, R120 22Ω

4 ETH_HSOUT
5 ETH_HSIN
R121 12.1kΩ

OSC1, OSC2, 25MHz, Y4
C82 27pF, C83 27pF

Y4 25 MHz, CL 18 pF: C82 = C83 = 2 x (18 − 4) = 28 -> 27 pF C0G 50 V.
APPROVED (Y4 / C3, 2026-10-25): calculation and parts.

ETH_VDD_IO_3V3
C85 100nF, R85 49.9Ω, R86 49.9Ω, R90 49.9Ω, R91 49.9Ω, C86 100nF

RX_N, RX_P, TX_N, TX_P

D27 ESD7104MUTAG

L2 DLW43MH201XK2L
L3 DLW43MH201XK2L

C87 100nF ISOLATION, C89 100nF ISOLATION
C90 100nF ISOLATION, C91 100nF ISOLATION

PHY_RX_N, PHY_RX_P, PHY_TX_N, PHY_TX_P

CN7
SM04B-GHS-TB
ETH

D11 LED Blue ETH Link/Act
LTST-S270TBKT
D10 LED
R83 1kΩ, R82 1kΩ, R84 10kΩ

ETH_VDD_IO_3V3
LTST-S270KRKT
D31 LED Red ETH 3V3 POWER
R154 1kΩ

MountingHole M3 x4 BASEBOARD HOLES

H205, H206, H207, H208

H205–H208: M3 baseboard holes, plain (no standoff):
3.1 mm drill, 5.0 mm plated pad to GND.

CARRIER EEPROM EXT SENSOR CAL.

VDD_3V3_AUX

U21
AT24C64D-MAHM-T

3..14 I2C3_SCL_BASE_MS5611_BARBED_EXTERNAL1
3..14 I2C3_SDA_BASE_MS5611_BARBED_EXTERNAL1

VCC A0, SCL A1, SDA A2, WP EP, VSS

EEPROM_A0, EEPROM_A1, EEPROM_A2
R153 10kΩ, R151 10kΩ, R152 10kΩ
0X51 Addressed

C114 1uF 16V, C115 100nF 16V

VBUS SENS CIRCUIT

VBUS, R115 10kΩ, R119 10kΩ

U22
MC74VHC1G32DFT1G
INB, INA, GND, VCC, OUTY

U26
MC74VHC1G09DFT1G
INB, INA, GND, VCC, OUTY

VDD_3V3_AUX, VDD_5V_SYS, R123 1kΩ

12,13 FMU_BOOTLOADER, TP18
VBUS_SENSE 3..12, TP22

C132 100nF, C136 100nF

PWR_FLAG (ERC drivers for the rails on this sheet)
PWR_FLAG
ETH_VDD_IO_3V3

RC INVERT BUFFER

VDD_3V3_IO, C203 100nF

U201
74LVC2G240DP
INVERT BUFFER

1OE, 1A, 2Y, GND, VCC, 2OE, 1Y, 2A

12,14 RC_1A
12,14 RC_2Y
RC_OE2, R203 220
NSBUS_OUT_EN 11,12
IO_SBUS_INPUT 11,12
IO_SBUS_OUTPUT 11,12
R204 10k

NRST BUFFER

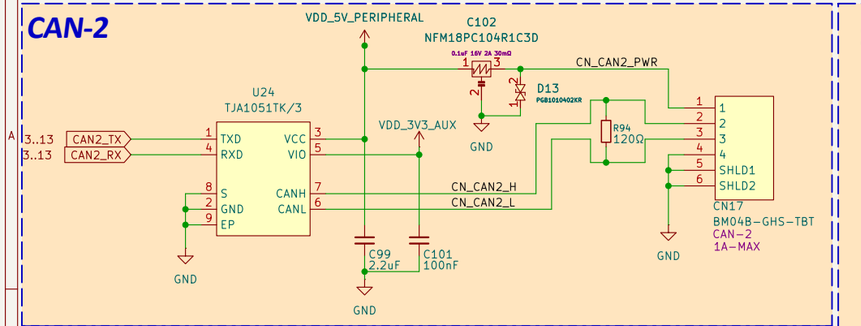
VDD_3V3_AUX, C208 100nF

U202
74LVC1G38
(NAND, open drain)

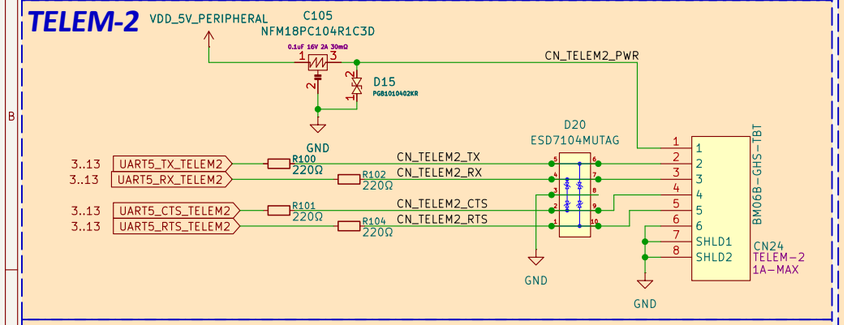
3..13 nARMED
12,13 FMU_RST_REQ
FMU_nRST 3..12

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VERO	Page	CARRIER_ELEMENTS
Drawn	Ahmet Vapurcu		
Reviewed			
	Naspier Core		
	Version: REV0 VERO	Size: A3	Page 12 Total 16
	VAP-SAN		

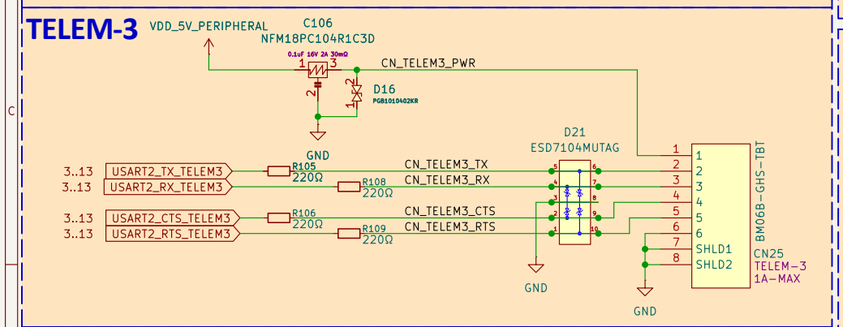
CAN-2



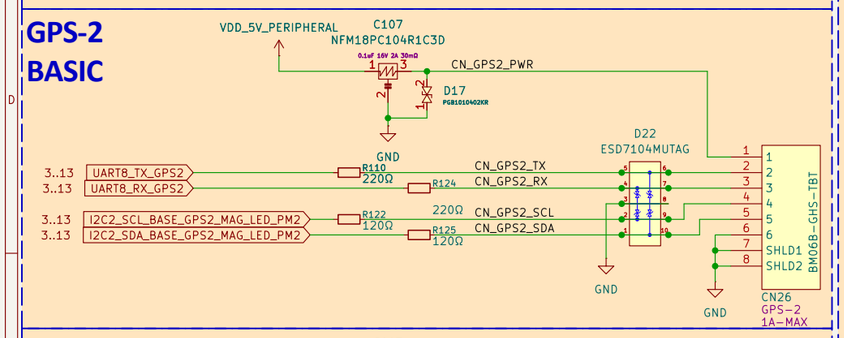
TELEM-2



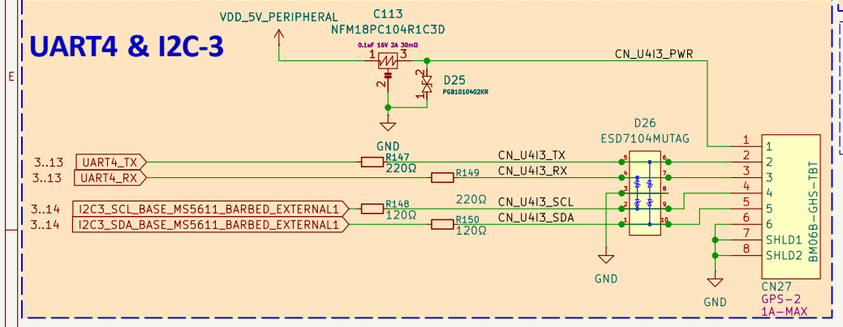
TELEM-3



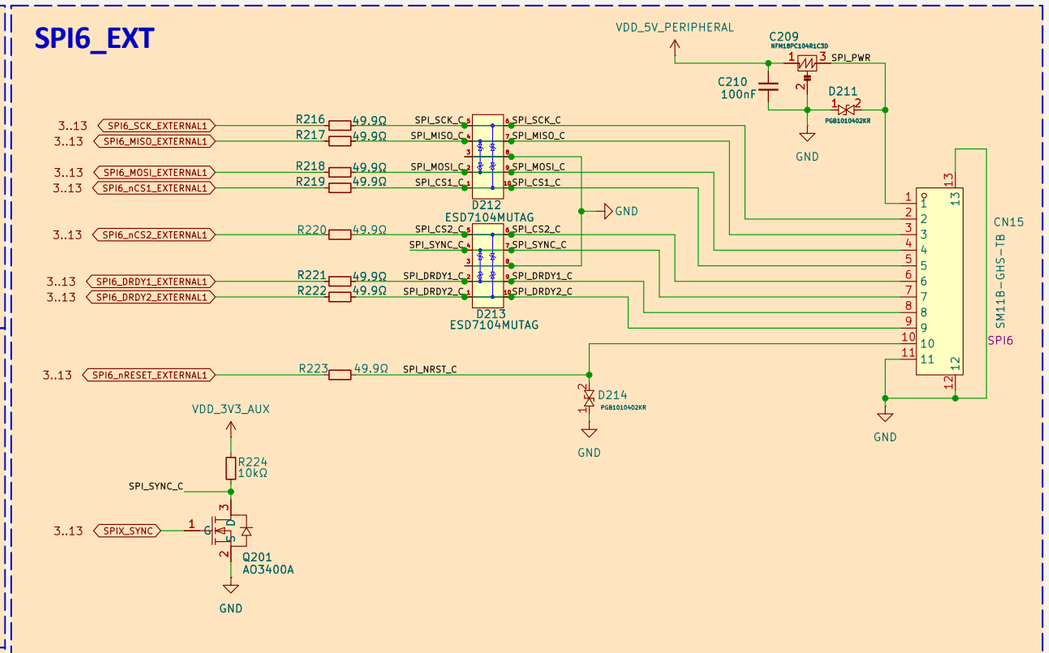
GPS-2 BASIC



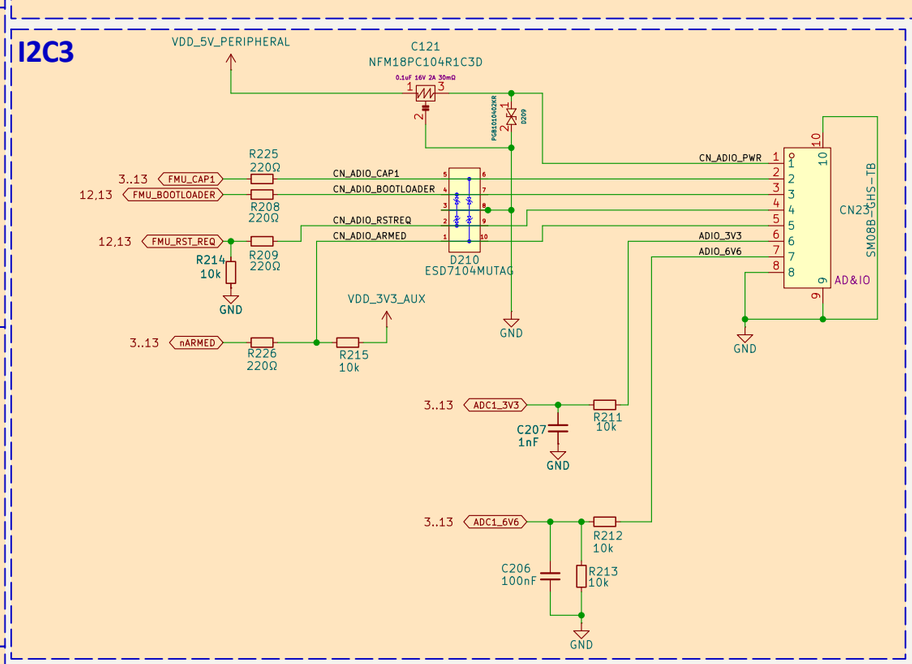
UART4 & I2C-3



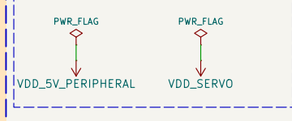
SPI6_EXT



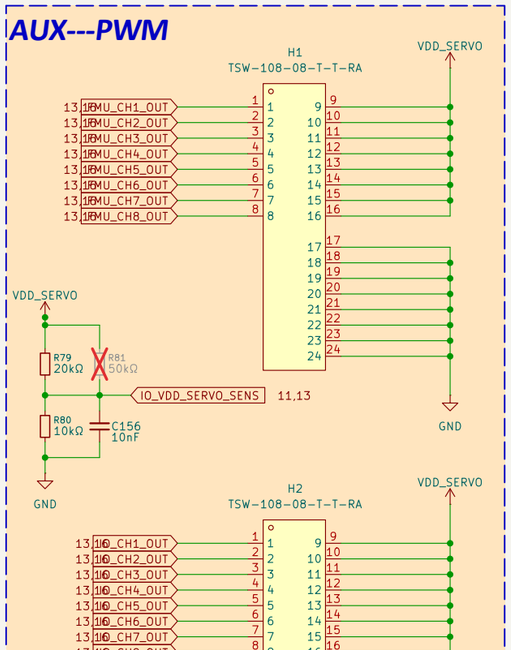
I2C3



PWR_FLAG (ERC drivers for the rails on this sheet)



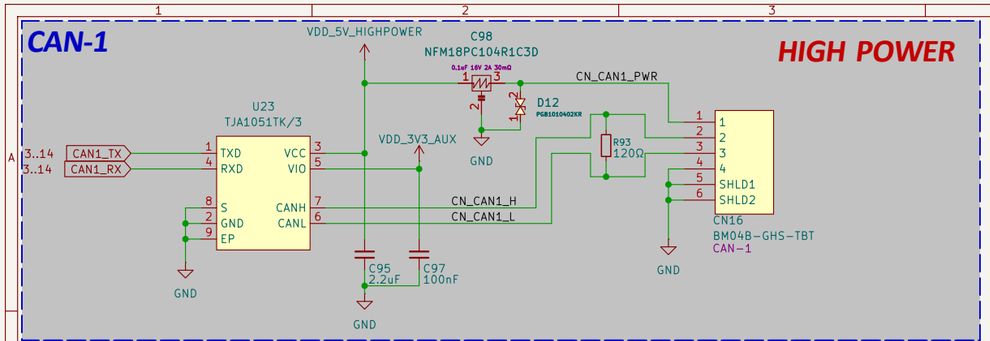
AUX---PWM

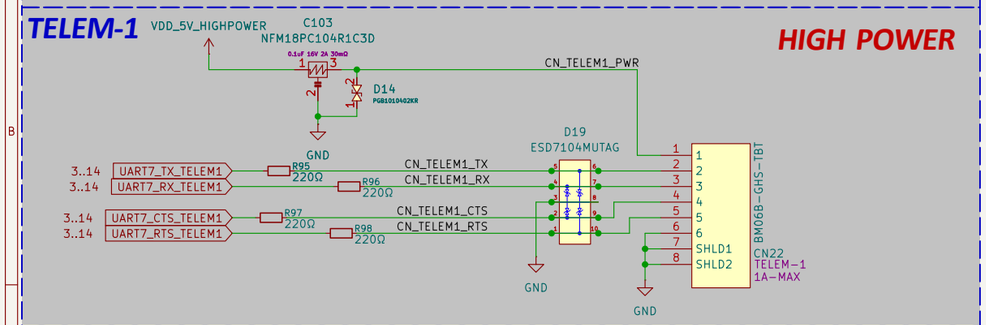


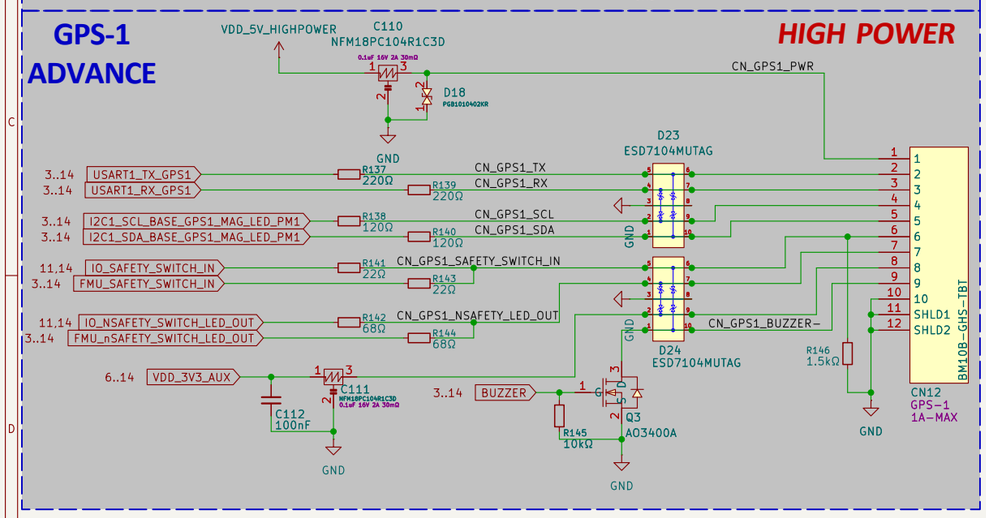
MAIN---PWM

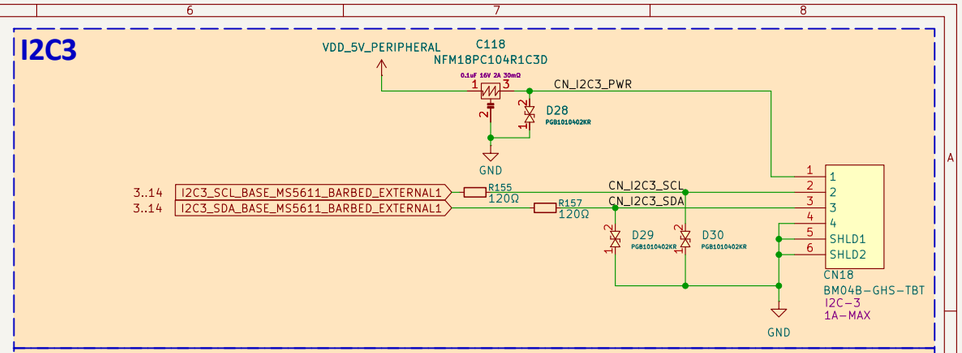
C5 (2026-10-25): H1 / H2 now on FMU_CHx_OUT / IO_CHx_OUT.
220 R + ESD7104 on all 16, IO channels level shifted to PWM_VDD: sheet CARRIER_PWM.

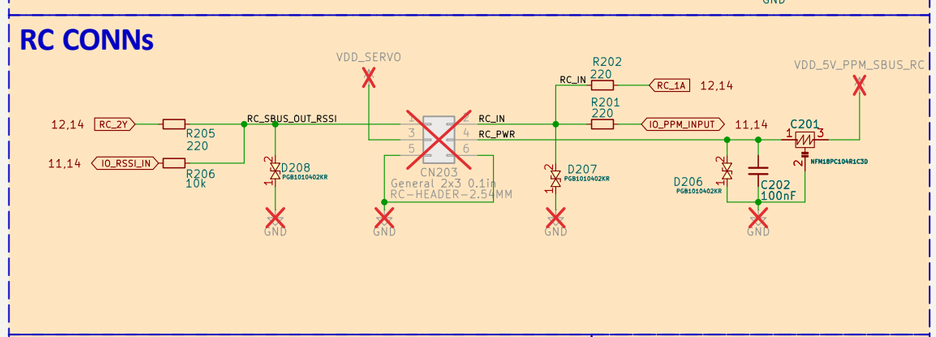
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_CONN_1**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version: REV0 VER0	Size: A3
		Page 13 Total 16	VAP-SAN

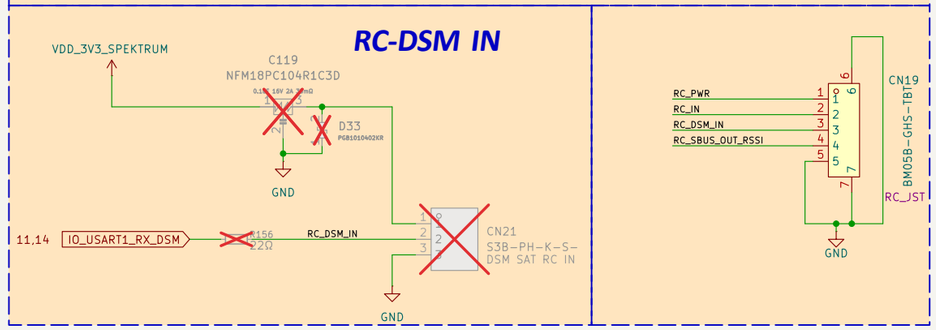




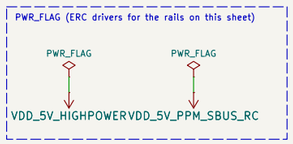






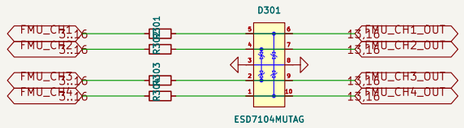


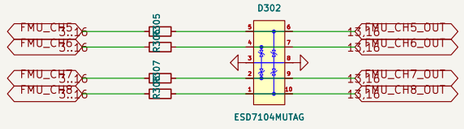
C4: CN203 (2x3 0.1 in header) stays DNP as an option.
R201 R202 R205 R206 C201 C202 D206–D208 are fitted: they also serve CN19 (GH RC IN).



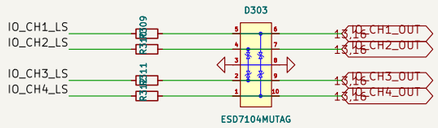
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_CONN_2**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version: REV0 VER0	Size: A3
		Page 14 Total 16	VAP-SAN

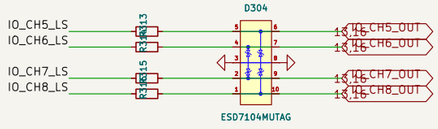
PWM OUTPUTS



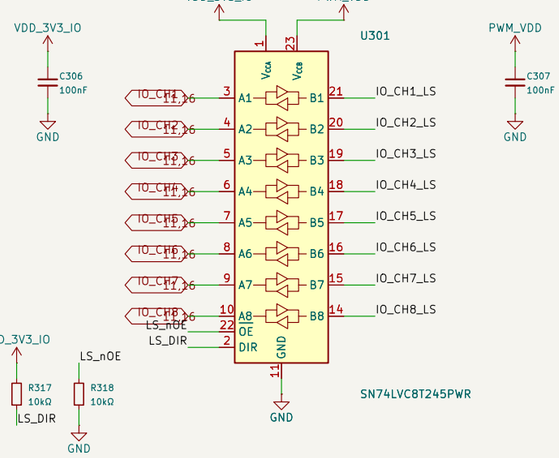


FMU_CH1–8: 220 R 1 % series + ESD7104 (RC05 page 9).
To H1 (MAIN PWM) as FMU_CHx_OUT.

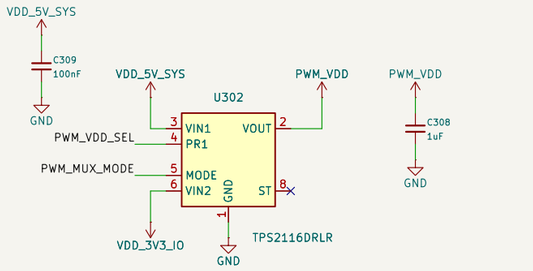


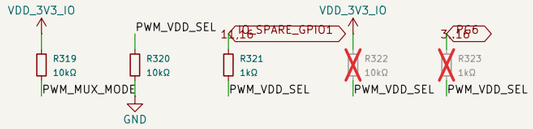


IO_CH1–8: level shifter U301 –> 220 R 1 % + ESD7104 (PAB–BASE V2 page 9).
To H2 (AUX PWM) as IO_CHx_OUT, at PWM_VDD (3.3 V or 5 V).



U301 SN74LVC8T245: A = VDD_3V3_IO (IO MCU), B = PWM_VDD.
DIR high (R317) = A –> B; -OE low (R318) = always enabled.

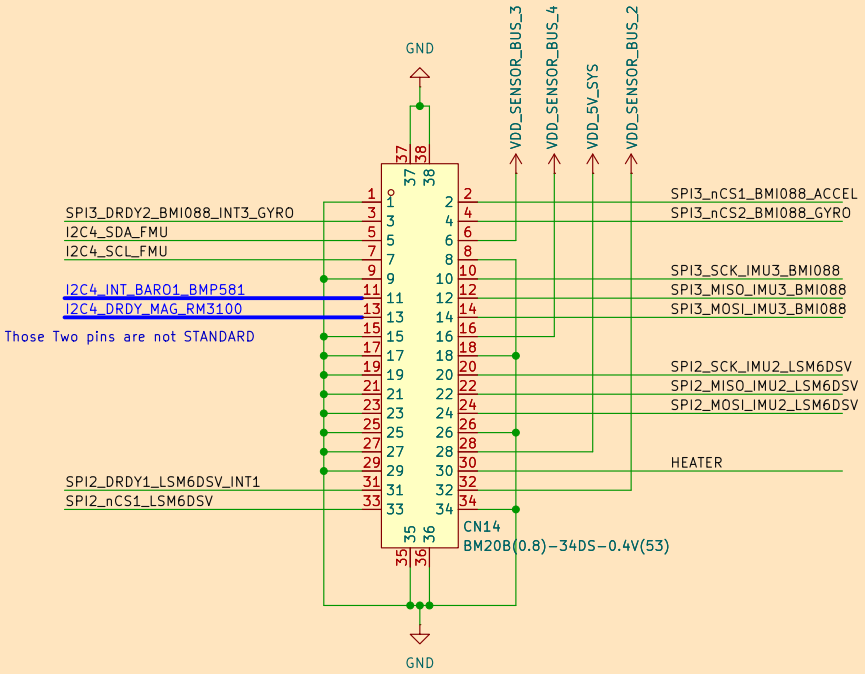


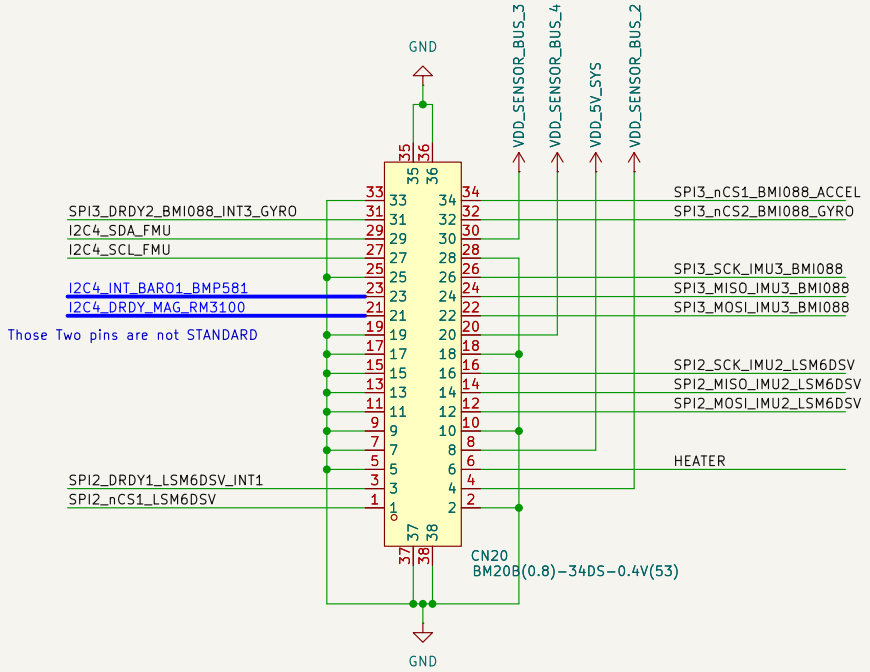


PWM_VDD select (as PAB–BASE V2 U27): U302 manual mode (MODE high via R319).
PWM_VDD_SEL high –> VIN1 VDD_5V_SYS (5 V PWM); low –> VIN2 VDD_3V3_IO (3.3 V PWM, default: R320 pull-down).
Driven by the IO MCU (IO_SPARE_GPIO1, PB6) through R321. Options DNP: R322 pull-up (fixed 5 V), R323 from FMU PG6.
CHECK: IO firmware must drive PB6 to pick 5 V; FMU and IO must not both drive the line (fit R321 or R323).

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-25
Board	NASPIER_CORE_REV0_VER0	Page	**CARRIER_PWM**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 16 Total 16
		REV0 VER0	A3	VAP–SAN

PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT





Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-15
Board	NASPIER_CORE_REV0_VER0	Page	
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version / Size / Page 15 Total 16	
		REV0 VER0 / A4 / VAP-SAN	

POWER SECTIONS Sheet

VDD_5V_SYS
VDD_5V_FILTERED
VDD_SENSOR_BUS_1
VDD_SENSOR_BUS_2
VDD_SENSOR_BUS_3
VDD_SENSOR_BUS_4
VDD_REG_FMU_3V3
VDD_3V3_AUX
FMU_RTC_3V3
VDD_5V_HIPOWER
VDD_5V_PERIPH
VDD_3V3_SPEKTRUM
VDD_SERVO

Power Buses

VDD_3V3_SENSORS1_EN
VDD_3V3_SENSORS2_EN
VDD_3V3_SENSORS3_EN
VDD_3V3_SENSORS4_EN
VDD_3V3_SD_CARD_EN
ETH_POWER_EN
VDD_5V_HIPOWER_nEN
VDD_5V_PERIPH_nEN
VDD_3V3_SPEKTRUM_POWER_EN

Power Buses Enable Lines

SCALED_VDD_3V3_SENSORS1
SCALED_VDD_3V3_SENSORS2
SCALED_VDD_3V3_SENSORS3
SCALED_VDD_3V3_SENSORS4
SCALED_V5
nPOWER_IN_A
nPOWER_IN_B
nPOWER_IN_C
VDD_5V_HIPOWER_nOC
VDD_5V_PERIPH_nOC
VDD_SERVO_SENS

Power Buses Feedback Lines

Dosya: Power_Sections_sch.kicad_sch

2026-09-20

Ahmet Vapurcu
NASPIER_CORE_REV0_VER0
VAP-SAN

Sheet: /
File: system-explain.kicad_sch

Title:

Size: A4	Date: 2026-10-15	**Rev: REV0 VER0**
KiCad E.D.A. 10.0.6		Id: 18/2

2026-09-20

Ahmet Vapurcu
NASPIER_CORE_REV0_VER0
VAP-SAN

Sheet: /POWER SECTIONS Sheet/
File: Power_Sections_sch.kicad_sch

Title:

Size: A4	Date: 2026-10-15	**Rev: REV0 VER0**
KiCad E.D.A. 10.0.6		Id: 1/2